Full Revision Booklet

Worked Solutions

Combined and renumbered from Practice Papers A and B

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1. Linear Programming

Source: Practice Paper A, original Question 1.

Let D be desks and S be shelving units.

(a) A suitable model is

$$\max Z = 58D + 46S$$

subject to

$$5D + 3S \leq 120$$

$$2D + 4S \leq 96$$

$$S \geq 8$$

$$D, S \geq 0$$

(b) At the optimum, carpentry and finishing are both binding:

$$5D + 3S = 120, \qquad 2D + 4S = 96$$

Hence

$$D = \frac{96}{7} \approx 13.71, \qquad S = \frac{120}{7} \approx 17.14$$

Maximum contribution:

$$Z = 58D + 46S = 1584$$

(c) Binding constraints: carpentry and finishing.

Slack values:

$$5D + 3S = 120 \Rightarrow 0 \text{ carpentry slack}$$

$$2D + 4S = 96 \Rightarrow 0 \text{ finishing slack}$$

$$S - 8 = \frac{64}{7} \approx 9.14$$

So the minimum shelving requirement is not binding.

(d) If shelving contribution rises to £52, the objective becomes

$$\max Z = 58D + 52S$$

The optimal corner remains the same:

$$(D, S) = \left(\frac{96}{7}, \frac{120}{7}\right)$$

with revised contribution

$$Z = \frac{11808}{7} \approx 1686.86$$

So the basis does not change.

(e) The new condition is

$$S \geq 0.3(D + S)$$

which simplifies to

$$3D - 7S \leq 0$$

At the original optimum,

$$\frac{S}{D+S} = \frac{17.14}{30.86} \approx 55.6\%$$

so the original solution remains feasible. The optimal plan is unchanged.

2. Binary Integer Programming

Source: Practice Paper A, original Question 2.

Let $x_A, \ldots, x_F \in \{0, 1\}$ where $x_i = 1$ if project i is selected.

(a) A suitable model is

$$\max B = 14x_A + 17x_B + 11x_C + 20x_D + 9x_E + 15x_F$$

subject to the budget constraint

$$6x_A + 8x_B + 5x_C + 9x_D + 4x_E + 7x_F \leq 21$$

and the logical constraints

$$x_D \leq x_B$$

$$x_A + x_F \leq 1$$

$$x_C + x_E \geq 1$$

$$x_i \in \{0, 1\}$$

(b) The optimal solution is

$$x_B = x_D = x_E = 1, \qquad x_A = x_C = x_F = 0$$

Total cost:

$$8 + 9 + 4 = 21$$

Maximum benefit:

$$17 + 20 + 9 = 46$$

(c) Binary variables are needed because a project cannot be partly selected in any practical sense. A continuous LP could return fractional values such as "0.4 of project D", which would not be meaningful here.

(d) Add

$$x_A + x_B + x_C + x_D + x_E + x_F = 4$$

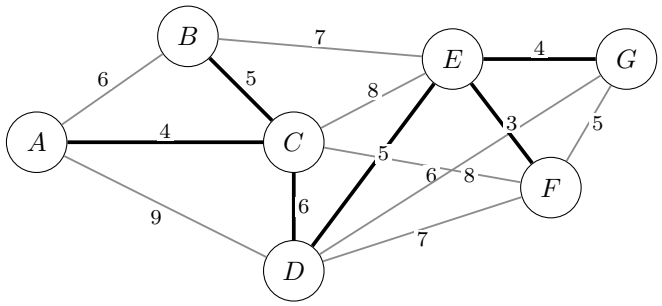
There is no feasible solution satisfying all rules, the budget and the requirement to choose exactly four projects.

(e) Example extension: allow projects to be staged over several years with different annual budgets and inter-year dependencies.

3. Minimum Spanning Tree

Source: Practice Paper A, original Question 3.

(a) A minimum spanning tree connects all nodes at minimum total cost without forming cycles.



(b) One correct Kruskal solution selects:

$$EF(3),\ AC(4),\ EG(4),\ BC(5),\ DE(5),\ CD(6)$$

(c) Total installation cost:

$$3 + 4 + 4 + 5 + 5 + 6 = 27$$

(d) The cheapest extra non-tree link has cost 5, for example FG.

So one cheapest backup addition is FG, giving revised total cost

$$27 + 5 = 32$$

(e) Limitation: an MST gives no built-in redundancy, so a single link failure can disconnect part of the network.

4. Simulation

Source: Practice Paper A, original Question 4.

(a) A suitable simulation model would treat weekly deliveries and weekly demand as random inputs. State variables would include opening stock, ending stock and overflow stock. Key outputs would be average weekly cost, overflow frequency and the average size of overflow inventory.

(b) One valid 12-week simulation can be produced using seed 2026, giving:

Deliveries:

$$7, 8, 8, 9, 7, 7, 8, 8, 9, 8, 9, 9$$

Demands:

$$8, 7, 6, 7, 5, 8, 7, 8, 7, 7, 6, 6$$

Assume opening stock is 0 and ending stock carries forward week to week.

For capacity 7:

Week	Open	Delivered	Demand	Sales	End Stock	Overflow	Cost
1	0	7	8	7	0	0	126
2	0	8	7	7	1	0	126
3	1	8	6	6	3	0	126
4	3	9	7	7	5	0	126
5	5	7	5	5	7	0	126
6	7	7	8	8	6	0	126
7	6	8	7	7	7	0	126
8	7	8	8	8	7	0	126
9	7	9	7	7	9	2	186
10	9	8	7	7	10	3	216
11	10	9	6	6	13	6	306
12	13	9	6	6	16	9	396

(c) Total 12-week cost for capacity 7:

$$2112$$

Estimated average weekly cost:

$$\frac{2112}{12} = 176.0$$

(d) Using the same 12-week random stream:

Capacity	Total Cost	Average Weekly Cost
6	2106	175.5
7	2112	176.0
8	2208	184.0
9	2304	192.0

On this run, capacity 6 gives the lowest average weekly cost of £175.5. The other averages are £176.0 for capacity 7, £184.0 for capacity 8 and £192.0 for capacity 9. Therefore, 6 bays appears to be the optimal capacity from this simulation. Increasing capacity beyond 6 raises the fixed bay cost faster than it reduces overflow cost on this run.

(e) Two reasons to run a longer or repeated simulation:

The 12-week result may be strongly affected by random variation.

Longer runs or replications would give a more stable estimate of average cost and better evidence for comparing capacities.

5. Forecasting

Source: Practice Paper A, original Question 5.

(a) Using periods 1–12 only, centred 4-quarter moving averages are available for periods 3–10:

Period	Demand	CMA
3	570	602.25
4	690	612.25
5	548	622.25
6	662	633.50
7	608	645.25
8	742	656.25
9	590	666.75
10	708	677.75

A linear trend fitted to these centred moving averages is

$$\hat{T}_t = 568.75 + 10.89t$$

where $t = 1$ corresponds to 2020 Q1.

(b) Seasonal indices:

Additive:

$$Q1 = -75.72, \quad Q2 = 29.16, \quad Q3 = -34.97, \quad Q4 = 81.53$$

Multiplicative:

$$Q1 = 0.8826, \quad Q2 = 1.0446, \quad Q3 = 0.9442, \quad Q4 = 1.1286$$

(c) Validation forecasts for 2023:

Quarter	Actual	Additive Forecast	Multiplicative Forecast
2023 Q1	636	634.60	626.93
2023 Q2	760	750.36	753.38
2023 Q3	692	697.13	691.23
2023 Q4	852	824.52	838.54

Using MAD:

$$\text{MAD}_{\text{add}} \approx 10.91, \qquad \text{MAD}_{\text{mult}} \approx 7.48$$

where absolute error is calculated as $|\text{Actual} - \text{Forecast}|$.

For the multiplicative model:

$$|636-626.93| = 9.07, \quad |760-753.38| = 6.62, \quad |692-691.23| = 0.77, \quad |852-838.54| = 13.46$$

so

$$\text{MAD}_{\text{mult}} = \frac{9.07 + 6.62 + 0.77 + 13.46}{4} \approx 7.48$$

For the additive model:

$$|636-634.60| = 1.40, \quad |760-750.36| = 9.64, \quad |692-697.13| = 5.13, \quad |852-824.52| = 27.48$$

so

$$\text{MAD}_{\text{add}} = \frac{1.40 + 9.64 + 5.13 + 27.48}{4} \approx 10.91$$

The multiplicative model is better.

(d) Forecasts for 2024 using the multiplicative model:

Quarter	Forecast
2024 Q1	665.38
2024 Q2	798.88
2024 Q3	732.35
2024 Q4	887.70

6. Transhipment

Source: Practice Paper A, original Question 6.

(a) Use a network flow model with variables defined on the links of the network.

Let the node sets be:

$$S = \{D_1, D_2, D_3\}, \qquad H = \{H_1, H_2\}, \qquad C = \{C_1, C_2, C_3, C_4\}$$

and let the arc set be

$$A = \{(i,h) : i \in S,\ h \in H\} \cup \{(h,j) : h \in H,\ j \in C\}.$$

Let

$$x_{uv} = \text{flow on arc } (u,v) \text{ for each } (u,v) \in A$$

and let k_{uv} be the corresponding unit transportation cost.

Objective:

$$\min Z = \sum_{(u,v)\in A} k_{uv} x_{uv}$$

Supply constraints for each depot $i \in S$:

$$\sum_{h\in H} x_{ih} \le s_i$$

Demand constraints for each clinic $j \in C$:

$$\sum_{h\in H} x_{hj} = r_j$$

Flow-balance constraints for each hub $h \in H$:

$$\sum_{i\in S} x_{ih} - \sum_{j\in C} x_{hj} = 0$$

Non-negativity:

$$x_{uv} \ge 0 \qquad \text{for all } (u,v) \in A$$

This is the same network-flow style used in Pyomo when variables are indexed over arcs and flow conservation is written at each transhipment node.

(b) One optimal shipment plan is:

Link	Quantity	Unit Cost
$D_1 \to H_2$	85	7
$D_2 \to H_1$	60	4
$D_2 \to H_2$	30	6
$D_3 \to H_2$	110	3
$H_1 \to C_1$	60	4
$H_2 \to C_2$	70	4
$H_2 \to C_3$	80	3
$H_2 \to C_4$	75	5

Total transportation cost:

$$2480$$

Unused supply: 15 units at D_1.

(c) Revised model changes:

$$\text{flow through } H_1 \leq 130$$

and

$$x_{H_2,C_3} = 0.$$

One revised optimal plan is:

Link	Quantity	Unit Cost
$D_1 \to H_1$	40	5
$D_1 \to H_2$	45	7
$D_2 \to H_1$	90	4
$D_3 \to H_2$	110	3
$H_1 \to C_1$	50	4
$H_1 \to C_3$	80	5
$H_2 \to C_1$	10	6
$H_2 \to C_2$	70	4
$H_2 \to C_4$	75	5

Revised total cost:

$$2520$$

(d) With 20 extra units at H_1, the cost falls back to 2480. One optimal link-flow solution is:

Link	Quantity	Unit Cost
$D_1 \to H_1$	85	5
$D_2 \to H_1$	55	4
$D_2 \to H_2$	35	6
$D_3 \to H_2$	110	3
$H_1 \to C_1$	60	4
$H_1 \to C_3$	80	5
$H_2 \to C_2$	70	4
$H_2 \to C_4$	75	5

Saving:

$$2520 - 2480 = 40$$

Extra capacity cost:

$$50$$

So it is not worthwhile.

7. Scheduling

Source: Practice Paper B, original Question 1.

(a) Applying Johnson's Rule gives the sequence

$$J_4,\ J_2,\ J_5,\ J_1,\ J_3$$

Main steps:

Step	Smallest time remaining	Placement rule	Partial sequence
1	2 on J_4 for M_1	place first	$J_4, _, _, _, _$
2	3 on J_2 for M_1	next from front	$J_4, J_2, _, _, _$
3	4 on J_3 for M_2	place last	$J_4, J_2, _, _, J_3$
4	5 on J_1 for M_2	next from back	$J_4, J_2, _, J_1, J_3$
5	5 on J_5 for M_1	fill remaining front slot	J_4, J_2, J_5, J_1, J_3

(b) Gantt timings:

Job	Machine 1	Machine 2
J_4	0–2	2–11
J_2	2–5	11–19
J_5	5–10	19–25
J_1	10–17	25–30
J_3	17–23	30–34

Minimum makespan:

$$34$$

(c) The original order J_1, J_2, J_3, J_4, J_5 gives makespan 39.
So Johnson's Rule improves the schedule by

$$39 - 34 = 5$$

time units.

(d) Solving the job shop problem gives an optimal schedule with makespan

$$19$$

One optimal schedule is:

Job	Task	Machine	Start	End
A	1	M_1	0	4
A	2	M_3	5	10
A	3	M_2	13	16
B	1	M_2	0	3
B	2	M_1	6	10
B	3	M_3	10	16
C	1	M_3	0	5
C	2	M_2	5	7
C	3	M_1	10	14
D	1	M_1	4	6
D	2	M_2	7	13
D	3	M_3	16	19

This can be obtained using a job shop formulation or an OR-Tools interval-variable model with precedence constraints within each job and `NoOverlap` constraints on each machine.

(e) Add a due-date constraint for job C:

$$C_C \leq 15$$

where C_C is the completion time of job C.

In the schedule above, job C already finishes at time 14, so this additional constraint is satisfied and one optimal makespan-19 schedule remains feasible. In other instances, such a due-date constraint could force resequencing or even cause infeasibility.

8. Binary Integer Programming

Source: Practice Paper B, original Question 2.

(a) Let $x_i = 1$ if base i is opened and 0 otherwise. Let $y_j = 1$ if district j is covered and 0 otherwise.

$$\max Z = 9000y_1 + 12000y_2 + 14000y_3 + 10000y_4 + 11000y_5 + 8000y_6$$

subject to

$$14x_A + 11x_B + 13x_C + 10x_D + 12x_E + 9x_F \leq 31$$
$$x_A + x_B + x_C + x_D + x_E + x_F \leq 3$$
$$y_1 \leq x_A + x_B$$
$$y_2 \leq x_A + x_C + x_E$$
$$y_3 \leq x_B + x_C + x_F$$
$$y_4 \leq x_A + x_D + x_F$$
$$y_5 \leq x_C + x_D + x_E$$
$$y_6 \leq x_E + x_F$$
$$x_i, y_j \in \{0, 1\}$$

(b) The optimal solution is

$$x_A = x_C = 1, \qquad x_B = x_D = x_E = x_F = 0$$

Total cost:

$$14 + 13 = 27$$

Covered districts:

$$1, 2, 3, 4, 5$$

Total covered population:

$$9000 + 12000 + 14000 + 10000 + 11000 = 56000$$

(c) Yes. In the optimal solution above, district 6 remains uncovered. This is acceptable in a maximum-coverage model because the objective is to maximise covered population subject to tight resources. In a set covering model, complete coverage would be mandatory.

(d) Revised model changes:

District 6 must be covered:

$$y_6 = 1$$

or equivalently

$$x_E + x_F \geq 1$$

Bases B and F cannot both be chosen:

$$x_B + x_F \leq 1$$

Under these revised conditions, one optimal solution is

$$x_E = x_F = 1, \qquad x_A = x_B = x_C = x_D = 0$$

with covered districts $2, 3, 4, 5, 6$ and total covered population

$$12000 + 14000 + 10000 + 11000 + 8000 = 55000$$

There are multiple optimal solutions with the same covered population; for example, $x_C = x_F = 1$ is also optimal.

9. Simulation

Source: Practice Paper B, original Question 3.

(a) Main components:

Entities: students.

Resource: advisers.

Events: arrival, service start, service completion.

Stochastic inputs: interarrival and service times.

Measures: number served, average wait, maximum wait, percentage waiting less than 8 minutes.

(b) One run with seed 24 gave:

Measure	Result
Students served	41
Average wait	1.17 minutes
Maximum wait	10.32 minutes
% waiting less than 8 minutes	95.12%

(c) Peak-period results:

Advisers	Served	Avg wait	Max wait	% below 8 min
2	70	1.88	12.99	95.71%
3	73	0.32	3.22	100%
4	70	0.08	2.23	100%

From this run, the minimum staffing level meeting the target is 3 advisers.

(d) A `PriorityResource` can be used in place of a standard `Resource`. Each arriving student would be labelled priority or standard, for example with probability 0.10 of being priority, and then would request service with a smaller priority number if they are priority users.

In SimPy terms, the main change is from a standard request such as

```
with desk.request() as req:
```

to a priority request such as

```
with desk.request(priority=0) as req:
```

for priority users and a larger number such as `priority=1` for standard users.

This would usually reduce priority-user waits and increase the waits experienced by standard users.

(e) Replications are needed because a single run may be unusually good or bad. If the model represented a continuing system rather than a finite session, a warm-up period might also be appropriate.

10. Forecasting

Source: Practice Paper B, original Question 4.

(a) Using periods 1–12 only, centred 4-quarter moving averages are available for periods 3–10:

Period	Demand	CMA
3	470	491.250
4	560	499.500
5	450	507.750
6	536	516.500
7	500	525.875
8	600	535.125
9	485	544.375
10	575	554.375

A linear trend fitted to these centred moving averages is

$$\hat{T}_t = 463.21 + 9.02t$$

where $t = 1$ corresponds to 2019 Q1.

(b) Seasonal indices:

Additive:

$$Q1 = -58.72, \quad Q2 = 19.91, \quad Q3 = -23.72, \quad Q4 = 62.53$$

Multiplicative:

$$Q1 = 0.8884, \quad Q2 = 1.0372, \quad Q3 = 0.9535, \quad Q4 = 1.1209$$

(c) Validation forecasts for 2022:

Quarter	Actual	Additive Forecast	Multiplicative Forecast
2022 Q1	522	521.76	515.68
2022 Q2	618	609.41	611.44
2022 Q3	572	574.80	570.71
2022 Q4	694	670.07	680.99

Using MAD:

$$\text{MAD}_{\text{add}} \approx 8.89, \qquad \text{MAD}_{\text{mult}} \approx 6.80$$

where absolute error is calculated as $|\text{Actual} - \text{Forecast}|$.

For the multiplicative model:

$$|522-515.68| = 6.32, \quad |618-611.44| = 6.56, \quad |572-570.71| = 1.29, \quad |694-680.99| = 13.01$$

so

$$\text{MAD}_{\text{mult}} = \frac{6.32 + 6.56 + 1.29 + 13.01}{4} \approx 6.80$$

For the additive model:

$$|522-521.76| = 0.24, \quad |618-609.41| = 8.59, \quad |572-574.80| = 2.80, \quad |694-670.07| = 23.93$$

so

$$\text{MAD}_{\text{add}} = \frac{0.24 + 8.59 + 2.80 + 23.93}{4} \approx 8.89$$

The multiplicative model is better.

(d) Forecasts for 2023 using the multiplicative model:

Quarter	Forecast
2023 Q1	547.74
2023 Q2	648.86
2023 Q3	605.11
2023 Q4	721.43

11. Linear Programming

Source: Practice Paper B, original Question 5.

Let C be casings and I be inserts.

(a) A suitable model is

$$\max Z = 40C + 34I$$

subject to

$$4C + 2I \leq 100$$

$$C + 3I \leq 90$$

$$I \leq 24$$

$$C, I \geq 0$$

(b) The optimal solution is

$$C = 13, \qquad I = 24$$

Maximum contribution:

$$Z = 40(13) + 34(24) = 1336$$

(c) Binding constraints: moulding and insert demand cap.

Slack values:

$$4(13) + 2(24) = 100 \Rightarrow 0 \text{ moulding slack}$$

$$13 + 3(24) = 85 \Rightarrow 5 \text{ assembly hours unused}$$

$$24 - 24 = 0$$

(d) The new condition is

$$I \geq 0.25(C + I)$$

which simplifies to

$$C - 3I \leq 0$$

At the original optimum,

$$13 - 3(24) = -59 \leq 0$$

so the original solution remains feasible and optimal.

(e) With 12 extra assembly hours, the assembly constraint becomes less restrictive, but the current solution already has 5 hours of assembly slack and is limited by moulding plus the insert cap. The optimal plan therefore does not change.

Extra contribution gained:

$$0$$

Maximum cost of buying all 12 hours:

$$12 \times 3 = 36$$

So the extra hours should not be bought.

12. Transportation

Source: Practice Paper B, original Question 6.

(a) Let x_{ij} be the quantity shipped from warehouse i to store j.

A suitable model is

$$\min Z = \sum_i \sum_j c_{ij} x_{ij}$$

subject to the supply constraints for each warehouse i:

$$\sum_j x_{ij} = s_i$$

demand constraints for each store j:

$$\sum_i x_{ij} = d_j$$

and non-negativity:

$$x_{ij} \geq 0$$

(b) One optimal plan is:

Route	Quantity
$W_1 \to S_1$	20
$W_1 \to S_4$	60
$W_2 \to S_2$	60
$W_2 \to S_3$	40
$W_3 \to S_1$	50
$W_3 \to S_3$	40

Total transportation cost:

$$1560$$

(c) With $W_2 \not\to S_2$ and at least 20 units from W_1 to S_4, one optimal revised plan is:

Route	Quantity
$W_1 \to S_2$	60
$W_1 \to S_4$	20
$W_2 \to S_3$	60
$W_2 \to S_4$	40
$W_3 \to S_1$	70
$W_3 \to S_3$	20

Revised cost:

$$1680$$

Increase in cost:

$$1680 - 1560 = 120$$

(d) Example extension: add vehicle capacities or delivery time windows so the model reflects route feasibility as well as shipping cost.